Week 8 Homework

This is a preview of the published version of the quiz

Started: Jul 2 at 7:50am

Quiz Instructions

Question 1

1 pts

Lesson 5.28: TRUE or FALSE? The ROUTE module is found in the Advanced Transfer template and allows you to move entities from Station to Station.

- True
- False

Question 2

1 pts

Lesson 5.29: A Sequence must consist of unique visitation locations... no repeat visits!

- True
- False

Question 3

1 pts

Lesson 5.30: You could use the Advanced Set functionality to define a "Set of Sequences".

- True

- False

Question 4

1 pts

Lesson 5.30: The name of the automatically supplied ARENA attribute that stores a particular entity's sequence is Entity.Sequence (what a surprise)!

- True
- False

Question 5

1 pts

Lesson 5.31: Cell 3 actually had TWO servers -- an "old"/slow guy and a "younger"/fast guy.

- True
- False

Question 6

1 pts

Lesson 5.32: TRUE or FALSE? You SEIZE-DELAY-RELEASE a conveyor.

- True
- False

Question 7

1 pts

Lesson 5.32: An ENTER module is usually paired with what other module?

- DEPART
- DISPOSE
- TERMINATE
- LEAVE

Question 8

1 pts

(Lesson 6.1: Introduction to Uniform Random Numbers.) TRUE or FALSE? It's actually a good thing for you to be able to reproduce a sequence of PRNs, should you so desire.

- True
- False

Question 9

1 pts

(Lesson 6.2: Some Lousy Generators.) Consider von Neumann's mid-square PRN method, and suppose that $X_0 = 6632$. What is $R_3 = X_3/10000$? [Note: For purposes of this problem, treat all X_i's as if they were 8 digits, e.g., treat 123456 as if it were 00123456.]

- a. 9834
- b. 0.9834
- c. 50055625
- d. 556
- e. 0.0556

Question 10

1 pts

(Lesson 6.3: Linear Congruential Generators.) YES or NO? Does $X_i = (X_{i-1} + 12)\mathbf{mod}(13)$ have full period?

- True
- False

Question 11

1 pts

(Lesson 6.3: Linear Congruential Generators. Problem 7.1 from Law 2015). Consider the generator

$X_i = (5X_{i-1} + 3)\mathbf{mod}(16)$.

Starting from $X_0 = 7$, find X_{500}.

- a. 0
- b. 6
- c. 7
- d. 11
- e. 38

Question 12

1 pts

(Lesson 6.3: Linear Congruential Generators.) Which uniform generator was recommended in class, at least as a desert island" generator?

- a. $X_i = 16807X_{i-1}\mathbf{mod}(2^{31})$

- b. $X_i = 16807 X_{i-1} \mathbf{mod}(2^{31} - 1)$
- c. $X_i = 16807 (X_{i-1} - 1) \mathbf{mod}(2^{31})$
- d. $X_i = 16807 (X_{i-1} - 1) \mathbf{mod}(2^{31} - 1)$

Question 13

1 pts

(Lesson 6.4: Tausworthe Generators.) Suppose that a Tausworthe generator gave you the series of bits 1010101. If you use all 7 bits, what Unif(0,1) random number would that translate to?

- a. 0.3825
- b. 0.5
- c. 0.6641
- d. 0.9826

Question 14

1 pts

(Lesson 6.5: Generalizations of LCGs.) TRUE or FALSE? There are some great PRN generators out there with incredible cycle lengths $\approx 2^{191}$ and even 2^{19937}!

- True
- False

Question 15

1 pts

(Lesson 6.6: Choosing a Generator --- Theory.) Which of the following statements about the RANDU generator is true?

- a. Something just ain't right about that boy.
- b. The generator is given by $X_i = 65539 X_{i-1} \mathbf{mod}(2^{31})$
- c. The PRNs appear at first glance to be uniform, but funny things happen when you look at the plots of the PRNs in multiple dimensions.
- d. The PRNs are distributed on just 15 hyperplanes.
- e. All of the above.

Question 16

1 pts

(Lesson 6.7: Statistical Considerations - Intro.) Suppose the guy on trial is actually guilty but you incorrectly acquit him. So you've incorrectly accepted the null hypothesis of innocence. What type of error have you just made - Type I or Type II?

- a. Type I
- b. Type II

Question 17

1 pts

(Lesson 6.8: Goodness-of-Fit Tests.) Suppose we observe 1000 PRNs to obtain the following data.

$interval\ i$	$[0.00, 0.25)$	$[0.25, 0.50)$	$[0.50, 0.75)$	$[0.75, 1.0]$
$number\ observed$	240	255	243	262

Conduct a χ^2 goodness-of-fit test to see if these numbers are approximately Unif(0,1). Use level of significance $\alpha = 0.05$. Here are some table entries that you may need: $\chi^2_{0.05,3} = 7.81$, $\chi^2_{0.05,4} = 9.49$, and $\chi^2_{0.05,5} = 11.1$. ACCEPT or REJECT?

- a. Accept
- b. Reject

Question 18

1 pts

(Lesson 6.9: Runs Tests for Independence.) Consider the following $n = 30$ PRNs. (Read from left to right, and then down.)

$$\begin{array}{cccccccccc} 0.79 & 0.68 & 0.46 & 0.69 & 0.90 & 0.93 & 0.99 & 0.86 & 0.33 & 0.22 \\ 0.60 & 0.18 & 0.59 & 0.38 & 0.69 & 0.76 & 0.91 & 0.62 & 0.22 & 0.19 \\ 0.11 & 0.45 & 0.72 & 0.88 & 0.65 & 0.55 & 0.31 & 0.27 & 0.46 & 0.89 \end{array}$$

Let's conduct a runs *up and down* test to test H_0: the U_i's are independent with level α = 0.05. ACCEPT or REJECT?

- a. Accept
- b. Reject

Question 19

1 pts

(Lesson 6.9: Runs Tests for Independence.) Suppose that U_1, U_2, U_3 are i.i.d. Unif(0,1). Let's denote the number of runs up-and-down by X. Find the EXACT distribution of X.

- a. X ~ Unif(0,1)
- b. X~ Norm(0,1)
- c. $\Pr(X = 0) = 0.2, \Pr(X = 1) = 0.3, \Pr(X = 2) = 0.3, \Pr(X = 3) = 0.2$
- d. $\Pr(X = 1) = 0.5, \Pr(X = 2) = 0.5$
- e. $\Pr(X = 1) = 1/3,\ \Pr(X = 2) = 2/3$

Not saved

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